DATA 505 - Statistics Using R
Fall 2025

Instructor information: Andrew Bean, Ph.D. (Adjunct Professor, Department of Mathematics and Computer Science)
Lectures: Tuesday 6:00 - 8:45 pm, Brother's College 21
Office Hours: virtual and by appointment only via Google calendar
`https://calendar.app.google/z4xbpyyQzaEk5Fz66`

Course Description. Covers how to use real-world data and draw intelligent conclusions on it using univariate and bivariate descriptive statistics such as centrality, spread, and correlation. Students have the opportunity to learn how to use the R language in the R-studio environment to obtain representative samples from large datasets and use the information in such samples to draw inferences on the larger datasets they were taken from. Course topics include graphical and tabular presentation of data, descriptive summary statistics, manipulation of data, regression, probability distributions, t-tests, and one-way analysis of variance.

Course Prerequisites. Students are expected to have taken courses in programming and in statistics.

Learning Outcomes. By the end of the course, students will be able to:

1. Select, create, and interpret appropriate graphical displays, descriptive statistics, and inferential statistics. They will apply the concepts of robustness and model validation;
2. Analyze real data within authentic applications;
3. Assess the quality of a statistical argument;
4. Use the R programming language;
5. Interpret results in language that is both statistically correct and understandable by someone who has not taken a course in statistics;
6. Apply statistical reasoning in broad areas of application including the social, biological, and physical sciences.

Text. R for Data Science (2nd edition) by Garrett Grolemund and Hadley Wickham,`https://r4ds.hadley.nz/`. This book is available online for free. You can also purchase a hard copy if you prefer. Additional readings will be assigned as needed.

Course website. The course website is available at `https://andrew-bean.github.io/data505`. It contains all course materials, including lecture slides. It was built with R and Quarto (`https://quarto.org/`). Source code for the website and lecture slides is available via a [private Github repository](https://github.com/andrew-bean/data505-r-for-statistics). For access, create a Github account and send me your username.

Moodle. The course Moodle site will be used to post grades and announcements.

Laptop Computer and Statistical Software. We will be using the software package R and the R- studio environment. These are open-source and as such may be downloaded and installed for free. Please visit `https://posit.co/download/rstudio-desktop/` first to install R, then to install RStudio. Please see the startup document on Moodle for full instructions.

The installation of the software needs to be completed by the end of class Tuesday, August 26. Try to install the software as soon as you can, so you have time to get help, if necessary. If you have any trouble, please contact the University Technology Service Center Helpdesk. Contact information and hours may be found here: `https://drew.edu/about/technology-resources/university-technology/contact-us/`.

Please have your laptop with R and R-Studio ready to be opened and run in each class. They will be used for interactive lab assingments and other small coding tasks. Try to use the same computer for homework that you will use for quizzes and exams; if you do this, you will be much less likely to experience computer problems during the tests.

Should you have a problem with your computer, and it was purchased through Drew, take it immediately to the Helpdesk. When you hand in your laptop for repairs, you may request a free loaner. If you purchased your computer elsewhere, you need to arrange for the repair, but you may rent a loaner while your computer is being repaired. Details of the loaner policy may be found at `https://www.drew.edu/university-technology/policies/loaner-policy/`.

Attendance. Class attendance and participation are integral to the academic experience at Drew University. As such, you are expected to attend every class unless you have a legitimate excuse, such as illness, family emergency or participation in a college sponsored event. Attendance is mandatory for successful completion of the course, whether you are taking a class in person or online. Students who face challenges participating at the scheduled time should contact the instructor immediately to make alternate arrangements. For legitimate absences, to the extent possible, students should watch recordings, study the PowerPoint slides, and submit due assignments. Students are responsible for class attendance and for the prompt and regular performance of all assigned work. Faculty members are not obligated to review class material, give makeup examinations, or make special arrangements to accommodate absences.

Use of Zoom: This class is not designed to be virtual (e.g., group activities, data-analysis labs). For that reason, I will NOT be recording lectures. Please be aware that attending any class on Zoom will likely impact your learning in a negative way, and you may only do so with a legitimate reason and with prior permission.

Canceled Class. If class is canceled for any reason on an exam date, the exam will be held on the next class period.

Email and communication. You are expected to check your Drew email regularly. I will use your Drew email to communicate with you about the course. Likewise I will check mine, **however** please do not expect immediate responses from me, especially during working hours.

Office Hours. Office hours are by appointment only. Please use the Google calendar link above to schedule a time to meet with me. If you cannot find a time that works for you, please email me and we can make alternative arrangements.

Late work and Missed Assignments. <u>There are no make-ups in this class, even for excused absences</u>. Missed or late assignments will earn a zero. To account for unforseen circumstances that may result in such a zero, 1 lab and 1 quiz will be dropped from your grade. If you know in advance that you will miss a class, please let me know as soon as possible. If you know in advance that you will miss a quiz or exam and you let me know ahead of time, you may be able to take it early by making appropriate arrangements with me. Students unable to submit work on time as a result of a documented extended illness should contact their instructor and Dana Giroux in the Office of Accessibility Resources (OAR) to coordinate appropriate accommodations.

Homework and Quizzes. There will be a homework assignment for this course most weeks. These are mandatory. You are expected to complete and understand the homework. Your understanding of the homework will be assessed via in-class quizzes, which will be short and based on the homework.

Components of Your Grade:

15% Labs (approx 9 with 1 drop)

Labs will be in-class hands-on activities graded on a "satisfactory/not-satisfactory" scale. Completion is the number 1 criterion for "satisfactory", but .

10% Weekly Quizzes (approx 6 with 1 drop)

Homework will be assigned but ungraded. Each class will begin with a homework quiz which will be pen-paper questions based on the homework. These will be kept quite simple (provided you have completed and understood the homework).

15% Case studies (2)

These will be hands-on data-analysis assignments using R to address an realistic research question using data.

$30\% = 15\% \times 2$ Exams (2)

Each exam will have two parts (pen-paper and R-based).

30% Final Exam

The final exam will have two parts (pen-paper and R-based).

Collaboration. No collaboration is permitted on exams and quizzes. The only programs you are allowed to run during exams and quizzes are R and RStudio. You are not allowed to have any cell phone out during quizzes and cell phones and smart watches must be at the front of the room if you bring them to exams. You are not allowed to reference any papers, websites, or other people during exams and quizzes. You may discuss the homework assignments with other students, but you should make sure that you understand the responses. If you simply type in the answer that someone else tells you, you lose a valuable opportunity to test your understanding.

Final Exam Policy. If extenuating circumstances occur, students may submit a Final Exam Reschedule request for review by the Associate Provost. Students may not negotiate a make-up date directly with the course instructor. Students may request to reschedule an exam only under the following circumstances:

- Two final exams scheduled at the same time;
- Three final exams scheduled in one calendar day; one of the exams will be rescheduled at the convenience of the instructor and the student;
- Serious illness, or personal emergency; the student is required to present documentation to validate.

The final exam schedule is visible on the Registrar's website by the beginning of each semester (`https://drew.edu/academic/office-of-the-registrar/registration/final-exam-schedule/`). Students are expected to schedule travel plans to occur AFTER their final exams.

Grade Cutoffs: We will use the following scale in determining your course grade from your course average. The course grades are not curved. Final grades are assigned according to the standard scale: A = 92 to 100; A- = 90 to 92; B+ = 88 to 90; B = 82 to 88; B- = 80 to 82; C+ = 78 to 80; C = 72 to 78; C- = 70 to 72; F = 0 to 70 (if registered as an undergraduate: class D+ = 68 to 70; D = 62 to 68; D- = 60 to 62; F = 0 to 60).

Academic Accommodations. Your experience in this class is important to me. If you have already established accommodations with the Office of Accessibility Resources (OAR), please provide me with a copy of your accommodation letter at your earliest convenience so we can discuss your needs in this course.

If you have not yet established services through the Office of Accessibility Resources (OAR), but have a temporary health condition or permanent disability that requires accommodations (conditions include but not limited to; mental health, attention-related, learning, vision, hearing, physical or health impacts), you are encouraged to contact OAR. OAR offers resources and coordinates reasonable accommodations for students with disabilities and/or temporary health conditions.

Although a disclosure may take place at any time during the semester, students are encouraged to do so early in the semester, because, in general, accommodations are not implemented retroactively. Office of Accessibility Resources contact information:
Director-Dana Giroux
Location-Brothers College, Room 119B
Phone: 973-408-3962
Email: dgiroux@drew.edu, disabilityserv@drew.edu

Academic Integrity Policy. All students are required to uphold the highest academic standards. Any case of academic dishonesty will be dealt with according to the guidelines and procedures outlined in Drew University's "Standards of Academic Integrity: Guidelines and Procedures," which is located in the academic policies section of the catalog (`https://drew-caspersen-catalog.coursedog.com/academicpolicies#academic-integrity-policy1`).

Generative AI. I encourage you to explore Artificial Intelligence (AI) resources for learning purposes. When using AI, you must (1) fully document and cite all material (including any code, graph, formula, table, mathematical equation, etc.) that was not generated by you, (2) check information generated by AI and take full responsibility for its accuracy, and (3) be able to identify where and how you used any AI tools and how they contributed to your work. You are not allowed to use AI for any quizzes or exams. Use of any AI tools without permission is unacceptable and will be reported as an academic integrity violation. If you have any doubts about what is acceptable please discuss them with the professor.

Supporting Student Success, Center for Academic Excellence. All Drew students can access subject tutoring, writing support and academic coaching free of charge in the Center for Academic Excellence (CAE), located in the library. Seeking help through learning support resources in the CAE can help you achieve your academic goals. To access the appointment schedule, please visit `drew.mywconline.com` and follow the instructions on the landing page; if a tutor is not available, please submit the Tutor Request form (`https://docs.google.com/a/drew.edu/forms/d/e/1FAIpQLScnfIAm1a9rdfa2s_fjrdW6tVdYtn04biG72acpWT9HixYt1w/viewform`). For any other questions, email cae@drew.edu.

Preliminary Schedule (subject to change)

Date	Topics	Notes
8/26	Introduction & Getting started	Lab 1
9/2	R Basics (variables, types, functions)	HW-Quiz 1 / Lab 2
	Last day to drop without “W” is Monday, 9/8	
9/9	Descriptive statistics & Summarizing data	HW-Quiz 2 / Lab 3
9/16	Basic statistical inference	HW-Quiz 3 / Lab 4
9/23	Data manipulation & visualization	HW-Quiz 4 / Lab 5
9/30	**Exam 1**	
10/7	Regression & Case study 1	
10/14	NO CLASS (Fall Break)	
10/21	Control flow (if/for/while)	HW-Quiz 5 / Lab 6
10/28	Writing functions	HW-Quiz 6 / Lab 7
11/4	**Exam 2**	
	Last day to drop with “W” is Tuesday, 11/18	
11/11	Sampling distributions & CLT	Lab 8
11/18	Simulation & Case study 2	HW-Quiz 7
11/25	Bootstrap & Case study 2	HW-Quiz 8
12/2	Shiny (UI-based apps)	Case study 2 due / Lab 9

Based on the university schedule, our final exam is scheduled for **Tuesday, Dec 9, from 7:30 to 10:30 pm**.